

N O R T H E A S T L S O G

A Multi-Source FIA-Proxy Classification of Late-Successional and Old- Growth Forest

Maine, New Hampshire, Vermont, New York | Cardinal HPC | April 2026

A. Weiskittel | Center for Research on Sustainable Forests, University of Maine

github.com/hobros/lsg-ne

TL;DR — Five Findings

What the regional analysis tells us before we get into the methods

1

Maine has the lowest LSOG share in the Northeast

v5.1 design-based estimate: ME 14.1%, NH 31.2%, VT 28.8%, NY 27.2%

2

Northeast forests are aging — except in Maine

NH +11.1, VT +9.1, NY +8.9 percentage points per decade (1999 to 2023). ME +1.6 pp, borderline significant.

3

Public lands carry 1.5 to 3x the LSOG concentration of private lands

ME federal 36% vs private 14%. NY State+Local has more LS+OG (583 K ac) than ME has across all owners.

4

OG-class plots store 2.6 to 3.6x the live carbon of Not-LSOG plots

ME OG = 139 Mg C / ha. Direct climate-mandate relevance for LD 1529.

5

Validation is partial: LS+OG combined is policy-stable, OG alone is not

Cohen's kappa for v5.1 LS+OG vs Pelz-OG = 0.25 (fair). OG-only kappa = 0.04 (essentially random).

PART 01

Methods

Why This Matters

Old-growth forest is rare, valuable, and politically salient — but hard to measure

THE CONTEXT

- **LD 1529 (Maine, 2025)**

Directs DACF to produce a comprehensive LSOG conservation report by Nov 2026, incorporating ecological, carbon, and economic considerations.

- **Hagan et al. 2024 LiDAR mapping**

Identified ~165,000 ha of unmapped old forest in Maine's unorganized territories.

- **Thompson et al. 2026**

Pathways for protecting Maine's remaining old-growth on private commercial timberlands.

- **Pelz et al. 2023 (FEM 549)**

Official USFS old-growth methodology applied to FIA on National Forest System lands.

PROJECT GOALS

1. **Build a defensible FIA proxy**

calibrated against Hagan, ORNL, and Pelz benchmarks.

2. **Extend to NH, VT, NY**

for regional comparison across the Northeast.

3. **Quantify carbon and ownership**

for direct LD 1529 policy use.

4. **Produce wall-to-wall maps**

via TreeMap imputation of FIA plots.

v5.1 Classifier: Six Dimensions, /12 Score

Operational at github.com/holoros/lhog-ne (R/fia_lhog_analysis_v5.r)

1 Large-tree BA

BA in trees DBH ≥ 20 in

1pt: ≥ 40 ft²/ac
2pt: ≥ 80 ft²/ac

2 Stand maturity

STDAGE (or max DBH fallback)

1pt: ≥ 80 yr (or DBH ≥ 24 in)
2pt: ≥ 120 yr

3 Structure

TPA-weighted SD of DBH

1pt: ≥ 5 in
2pt: ≥ 8 in

4 Total stocking

Total live BA

1pt: ≥ 100 ft²/ac
2pt: ≥ 150 ft²/ac

5 Deadwood

Snag TPA, data-driven

1pt: ≥ 75 th percentile
2pt: ≥ 90 th

6 Canopy height

Potapov 2021 GEDI RH95

1pt: ≥ 18 m
2pt: ≥ 25 m

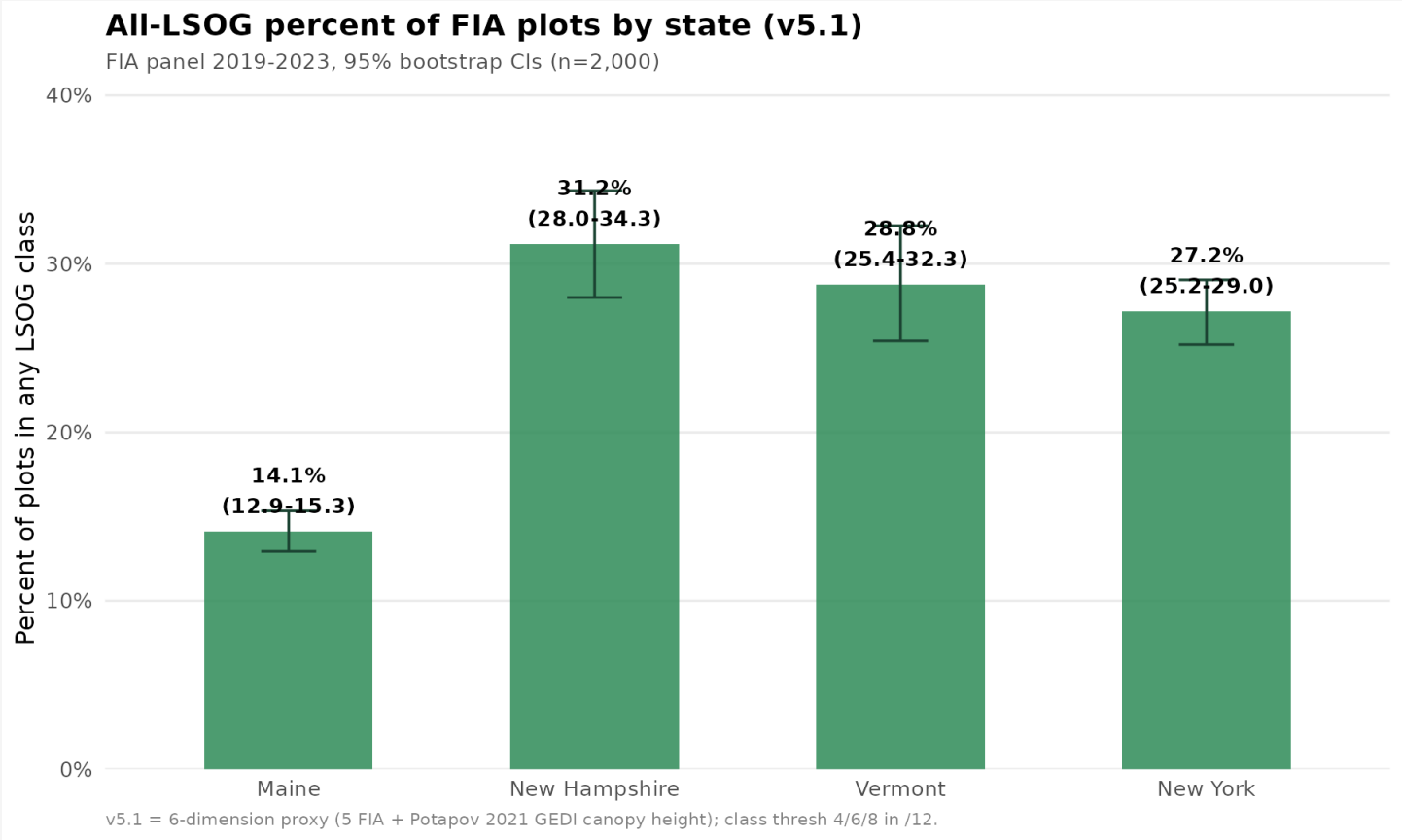
CLASS THRESHOLDS: Transitioning LS ≥ 4 | LS ≥ 6 | OG ≥ 8 (out of 12)

PART 02

Results

Headline Regional Results (FIA panel 2019-2023)

v5.1 plot-based bootstrap estimates with 95% confidence intervals



STATE-LEVEL SHARES

Maine

14.1%

any-LSOG (95% CI 12.9 - 15.3)

New Hampshire

31.2%

any-LSOG (95% CI 28.0 - 34.3)

Vermont

28.8%

any-LSOG (95% CI 25.4 - 32.3)

New York

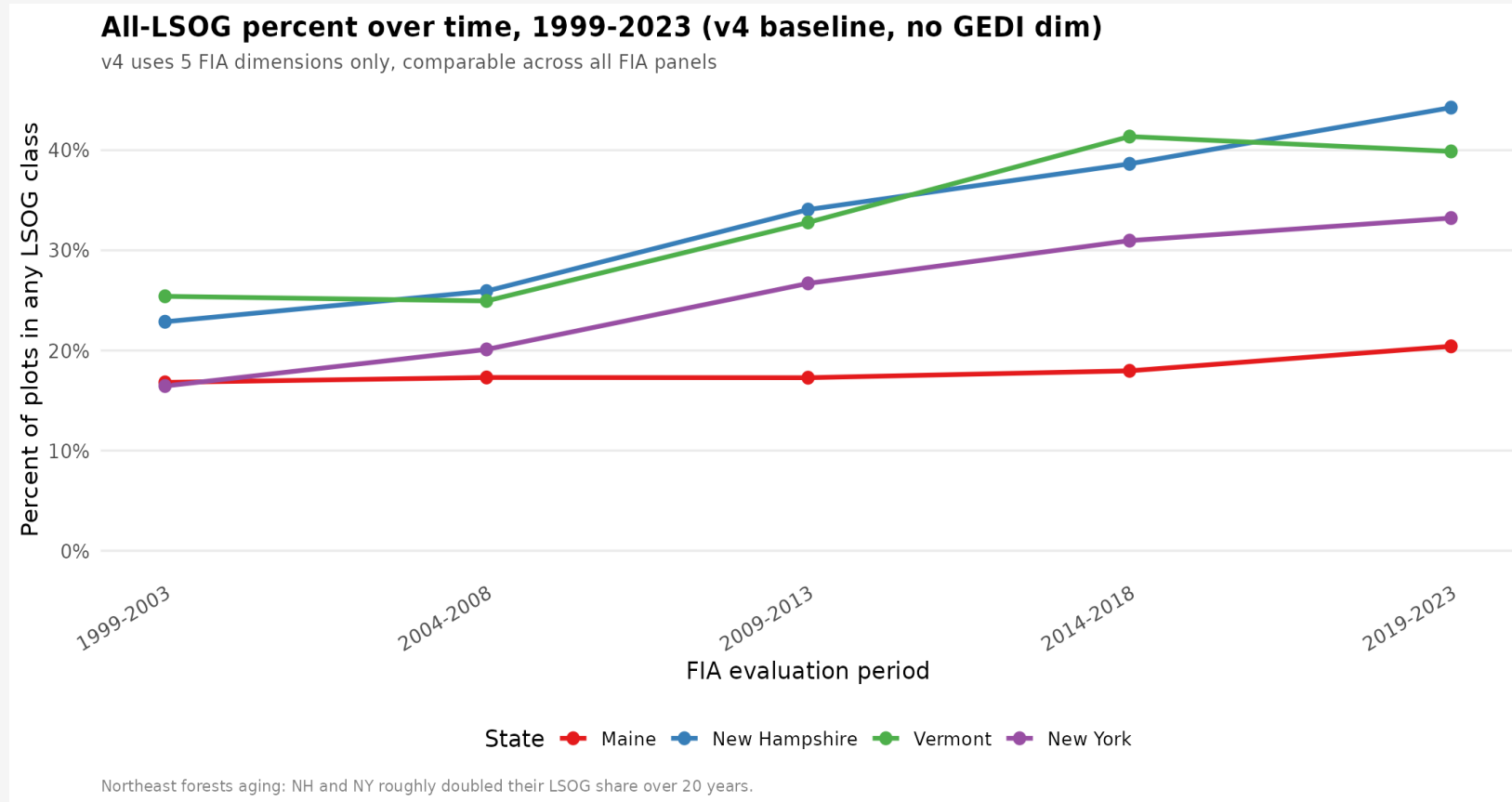
27.2%

any-LSOG (95% CI 25.2 - 29.0)

Maine = lowest LSOG share in NE
despite highest absolute forest area

Forests Are Aging: 20-year LSOG Trend (1999-2023)

v4 baseline (no GEDI), comparable across all 5 FIA panels



TREND TEST

All-LSOG % per decade

NH +11.1 pp

$p < 0.001$

VT +9.1 pp

$p = 0.024$

NY +8.9 pp

$p = 0.002$

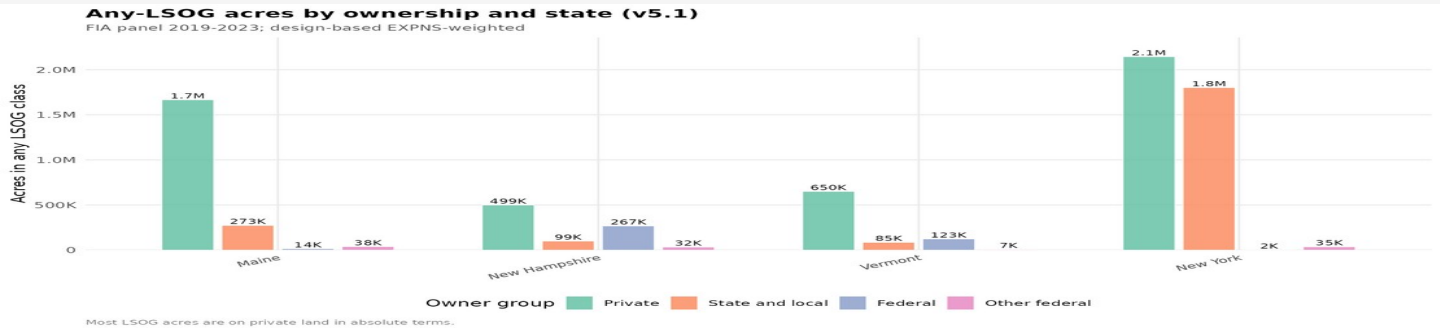
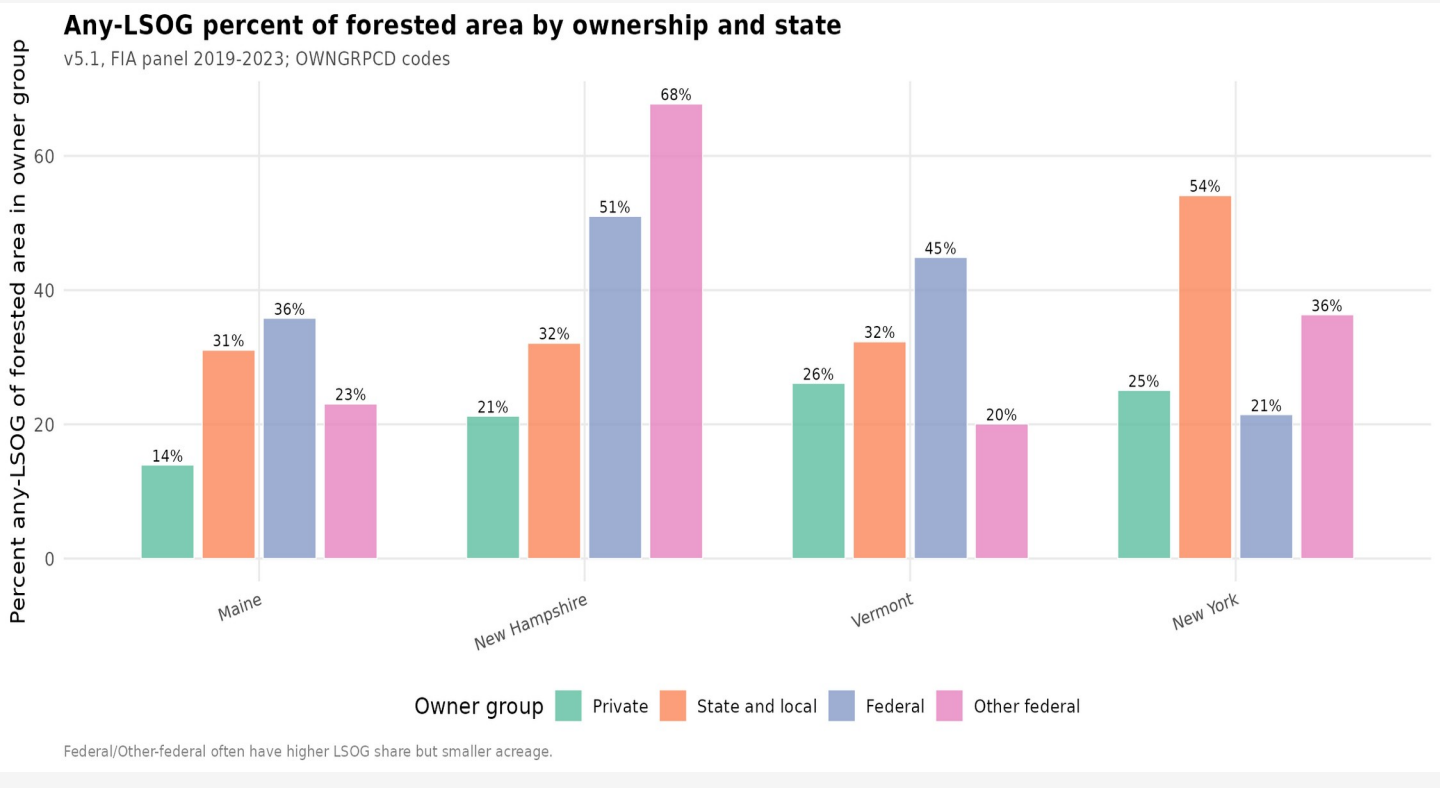
ME +1.6 pp

$p = 0.057$

Maine slope = borderline
(intensive harvest dampens aging signal)

Ownership: Public Lands Have 1.5-3× Higher LSOG Concentration

Design-based EXPNS-weighted estimation, latest panel



KEY POLICY FINDINGS

NY State+Local lands

→ 17.5% LS+OG = 583 K ac
More than ME has across all owners.

Public concentration

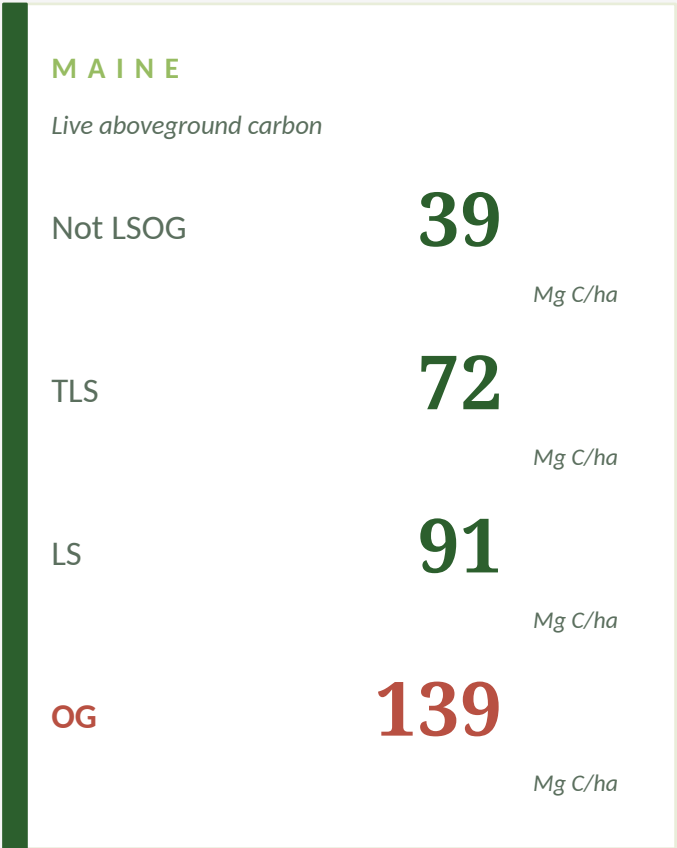
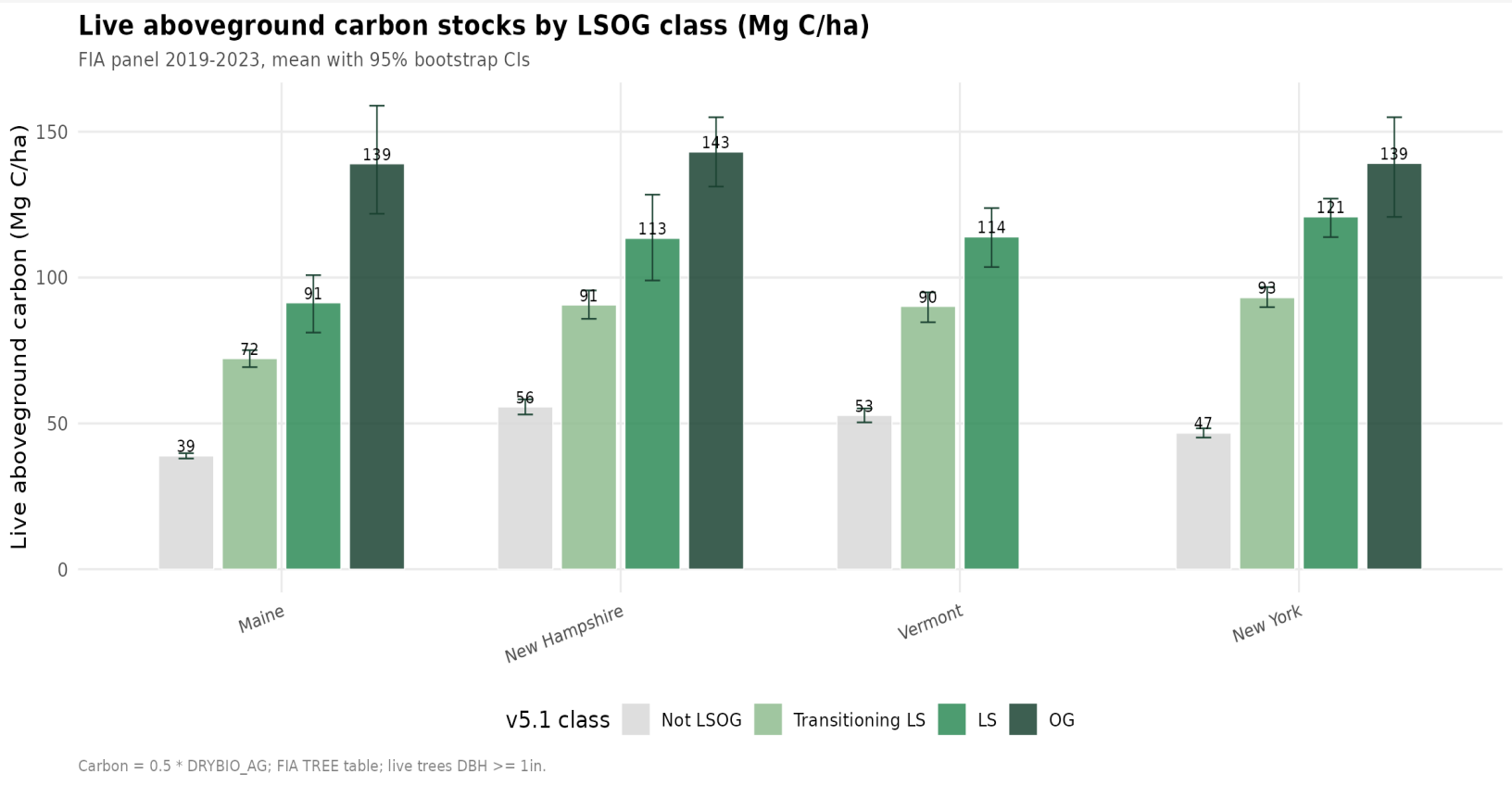
→ ME federal 36% vs private 14%
→ NH federal 51% vs private 21%
→ VT federal 45% vs private 26%

Where total acres live

Private LS+OG total ~987 K ac across NE
the natural target for PES / easements

Carbon: OG Plots Store 2.6-3.6× the Live Carbon of Not-LSOG

Direct LD 1529 climate-mandate relevance



Protecting Maine's ~24 K ac of OG class preserves ~100 Mg C/ha MORE carbon than letting it revert to Not-LSOG. Across the broader 1.99 M ac LSOG pool, that's substantial avoided emissions in policy-relevant terms.

PART 03

Validation and Outlook

Validation Against Independent Products

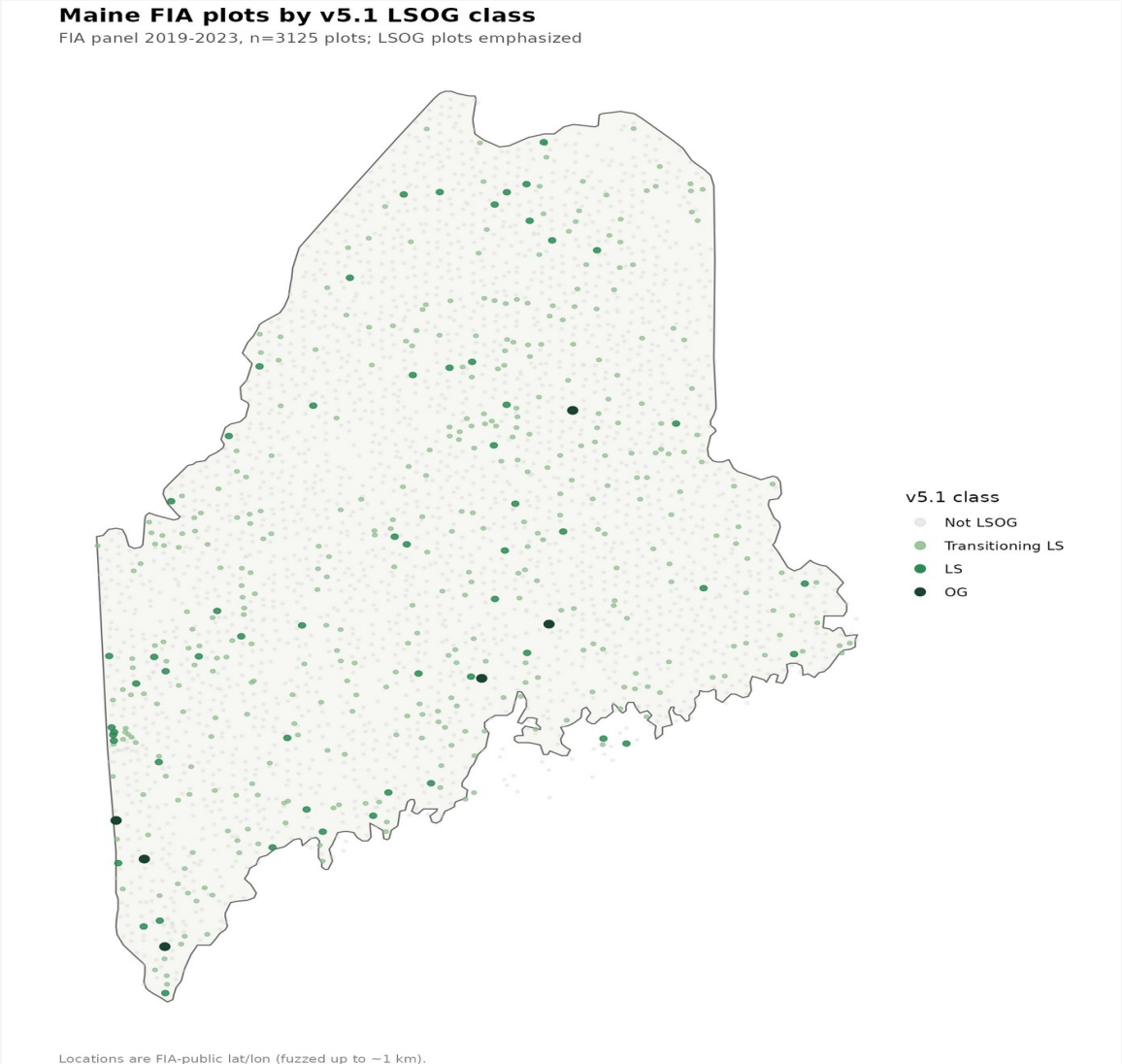
v5.1 fits within range of LiDAR, Bayesian, and USFS-official OG estimates

Source	Coverage	any-LSOG / OG share	Phase 7 kappa vs v5.1
v5.1 (this work)	All ME private + public	ME 14% any LSOG, 0.19% OG	—
Hagan et al. 2024 LiDAR	Maine UT only (9.5 M ac)	21.4% LSOG, 1.0% OG (95 K ac)	BLOCKED (raster pending)
Bruening 2026 (ORNL 2498)	Statewide MOG-prob > 50	~33% mature, 0.13% OG	kappa ~0 on OG-class
Pelz 2023 (FEM 549)	NE NFS lands (~1 M ac)	10.1% Pelz-OG of NFS plots	0.25 LS+OG vs Pelz-OG
TreeMap (this work, v2)	Maine wall-to-wall 30 m	8.0-8.2% any-LSOG	—

Take-away: v5.1 LS+OG combined has fair agreement ($\kappa \approx 0.25$) with Pelz's USFS-official OG criteria. The OG-specific class is poorly validated across all independent products ($\kappa \sim 0$). For policy reporting, LS+OG combined is the most stable category.

Wall-to-Wall: Maine 2020 / 2022 via TreeMap × FIA Imputation

Phase 8 v2 closes the Unknown coverage gap to 0.33% via 1999-2023 panel lookup



MAINE TREEMAP RESULTS

Year	Class	Acres	% pixels
2020	Trans LS	1.25 M	7.5%
2020	LS	118 K	0.7%
2020	Not LSOG	15.30 M	91.5%
2022	Trans LS	1.23 M	7.4%
2022	LS	113 K	0.7%
2022	Not LSOG	15.29 M	91.6%

Why ME 8% < v5.1 plot-based 14%

Phase 8 v2 uses v4-style scoring (no Potapov GEDI canopy height), since the Potapov product is single-year 2019 and not applicable to plots from older FIA panels. Phase 8 v3 (Potapov-aware wall-to-wall) is the natural next step.

Phase 7: Pelz 2023 Cross-Validation on NE NFS Lands

Independent test using USFS-official Eastern Region OG criteria

CONFUSION MATRIX (925 NFS plots in unified table)

v5.1 class	Not Pelz-OG	Pelz-OG	Total
Not LSOG	467	36	503
Transitioning LS	314	49	363
LS	31	24	55
OG	1	3	4
Total	813	112	925

COHEN'S KAPPA

v5.1 OG vs Pelz OG0.044

essentially random

v5.1 LS+OG vs Pelz OG0.253

fair agreement

v5.1 any-LSOG vs Pelz OG0.115

slight

OG-share estimates differ by 25× across products on NE NFS plots

v5.1

0.4%

Hagan UT

1.0%

Pelz R9

10.1%

Limitations & Caveats

Read every estimate alongside its weakness

1

OG-class precision is low

v5.1 OG kappa ≈ 0 vs Pelz, ORNL, and ranks. The four v5.1 OG plots may not match Pelz, ORNL, or Hagan OG calls. Don't over-interpret state-level OG percentages.

2

Hagan validation pending

Hagan 100m LiDAR raster is the canonical Maine UT old-growth reference. Phase 6 pipeline is fully scaffolded; awaits raster acquisition (Our Climate Common, LD 1529 working group, or Harvard Forest).

3

FIA plot fuzzing introduces ~30 m uncertainty

Public FIA lat/lon are fuzzed up to ~1 km. For Potapov RH95 extraction at 30 m pixels this introduces small but systematic noise. DUA true coordinates would tighten this.

4

TreeMap NE-extension blocked on TM_ID encoding

Phase 8 v2 ME results are wall-to-wall validated; NH/VT/NY extension hit a category-encoding mismatch in CONUS-cropped rasters. Three debug paths documented for next session.

5

ORNL 2498 trains on FIA labels

Bruening et al. 2026 use FIA MOG labels as response variable. Cross-validation is therefore a coherence check, not blind validation.

Project Status and Next Steps

33+ commits on github.com/holoros/lzog-ne; ready for Phase 6 the moment Hagan raster lands

DELIVERED

- v5.1 operational classifier
- Multi-state regional run (ME/NH/VT/NY)
- Bootstrap and FIA design-based CIs
- Ownership, carbon, forest-type breakdowns
- Pelz cross-validation (Phase 7)
- TreeMap NE wall-to-wall (Phase 8 v3)
- Master plot-level CSV (13,432 rows)

QUEUED UNBLOCKED

- Phase 8 v3-Potapov: GEDI-aware wall-to-wall
- State-specific RH95 thresholds
- Maine UT polygon-based filter
- rFIA design-package cross-validation
- GEDI L2A direct integration
- DUA true-coordinate refinement
- Manuscript v1 (intro drafted; methods next)

BLOCKED

- Phase 6 Hagan validation
Awaits 100 m LiDAR LSOG raster
- Acquisition routes:
 >> *Our Climate Common*
 >> *LD 1529 / DACF working group*
 >> *Harvard Forest Thompson team*